

Basic parametric tests

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- Parametric tests are the set of fundamental tests conducted when the data assume to come from a population with a known distribution, usually a normal distribution (i.e. normality assumption)
- Parametric tests require a numerical data
- There are 3 most common basic parametric tests:
 1. Independent t-test
 2. Paired t-test
 3. One-way ANOVA
- Parametric tests are quite robust to some degree of non-normality if the sample size is large and there is no influence of an outlier

Independent t-test

- Compares the two mean in relation to the variation in the data
- Other names: Student's t-test, two-sample t-test
- Independent t-test is conducted when:
 - The dependent variable is numerical
 - The independent/grouping variable is categorical (2 levels)
- Assumptions:
 1. Random sample
 2. The grouping and the outcome variables are independent of each other
 3. Normality - the outcome variable is normally distributed in each group
 4. Homoscedascity (equal variance) - the variance between groups is approximately normal

- Examples:
 - Comparing the systolic blood pressure between patients clinic A and clinic B
 - Comparing the BMI of doctors and pharmacists in a hospital

Independent t-test: Analysis

- Background: we want to compare the wound healing time between smoker and non-smoker
- Steps:
 1. State the hypothesis
 2. Check the assumptions
 3. Interpretation
 4. Conclusion

- Step 1:
 - H_0 : There is no difference in healing time between smokers and non-smokers
 - H_A : There is a difference in healing time between smokers and non-smokers

Packages

```
library(foreign)
```

```
library(knitr)
```

```
library(ggplot2)
```

```
library(dplyr)
```

Read the data

```
dat <- read.spss("Data/ind_t.sav", to.data.frame = TRUE)
```

First 6 rows of the data

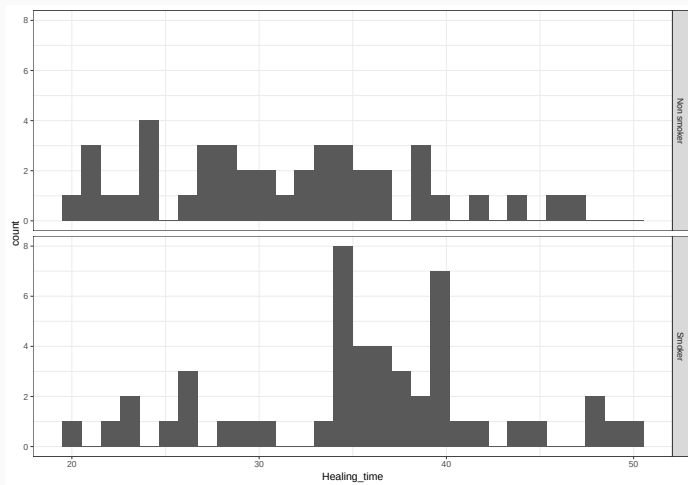
```
head(dat) %>% kable()
```

ID	Smoking	Healing_time
1	Smoker	44
2	Non smoker	33
3	Non smoker	28
4	Non smoker	42
5	Smoker	50
6	Non smoker	46

- Step 2:

1. Normality - does not looks normal, but, let's assume normal

```
ggplot(dat, aes(Healing_time)) +  
  geom_histogram() +  
  facet_grid(rows = vars(Smoking)) +  
  theme_bw()
```



- Step 2:
 2. Equal variance: $p = 0.979$ (equal variance is assumed)

```
var.test(Healing_time ~ Smoking, data = dat)
```

Ratio of Variances	F-statistic	p-value
1.006	1.006	0.979

- Step 3: Interpret:
 - $p \text{ value} < 0.05$
 - 95% CI does not cross 0
 - There is a significant difference in wound healing time between smoker and non-smoker

```
t.test(Healing_time ~ Smoking, data = dat)
```

t-statistic	df	p-value	Lower 95% CI	Upper 95% CI
-3	86.353	0.004	-7.526	-1.527

- Step 4: Conclusion:
 - Smoker had a significantly longer wound healing time compared to non-smoker

```
dat %>%  
  group_by(Smoking) %>%  
  summarise(mean = mean(Healing_time))
```

```
# A tibble: 2 x 2  
  Smoking      mean  
  <fct>      <dbl>  
1 Non smoker  31.3  
2 Smoker     35.8
```

Paired t-test

- Compares the two dependent or related samples:
 - Pre and post
 - Matched study design
 - Closely related subjects like twins
 - Cross over study design
- Other names: dependent sample t-test
- Paired t-test is conducted when:
 - The dependent variable is numerical
 - The groups are related

- Assumptions:
 1. Random sample
 2. Dependent/related sample
 3. Normality - the difference between the groups is normally distributed
- Examples:
 - Comparing the cholesterol level before and after 6 month of physical exercise
 - Comparing the effective of two drugs (A and B) in weight loss (cross over study design)

Paired t-test: Analysis

- Background: we want to compare the satisfaction level before and after the implementing the new system in a hospital
- Steps:
 1. State the hypothesis
 2. Check the assumptions
 3. Interpretation
 4. Conclusion

- Step 1:
 - H_0 : There is no difference in satisfaction level pre and post implementation of the new system
 - H_A : There is a difference in satisfaction level pre and post implementation of the new system

Packages

```
library(foreign)
```

```
library(knitr)
```

```
library(ggplot2)
```

Read the data

```
dat <- read.spss("Data/paired_t.sav", to.data.frame = TRUE)
```

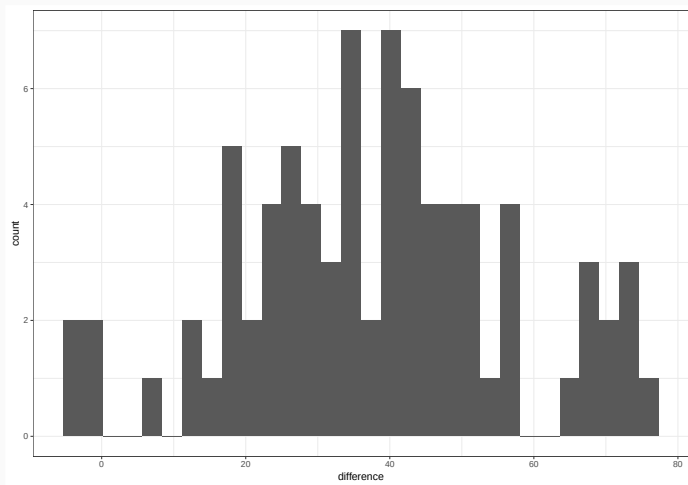
```
# First 6 rows of the data
```

```
head(dat) %>% kable()
```

ID	satisfaction_pre	satisfaction_post	difference
1	21	69	48
2	27	51	24
3	27	78	51
4	48	95	47
5	52	86	34
6	28	96	68

- Step 2:
 1. Normality - looks approximately normal

```
ggplot(dat, aes(difference)) +  
  geom_histogram() +  
  theme_bw()
```



- Step 3: Interpret:
 - $p \text{ value} < 0.05$
 - 95% CI does not cross 0
 - There is a significant difference in the satisfaction level pre and post implementation of the new system

```
with(dat, t.test(satisfaction_post, satisfaction_pre, paired = T))
```

t-statistic	df	p-value	Lower 95% CI	Upper 95% CI
17.934	79	0	33.593	41.982

- Step 4: Conclusion:
 - There is a significant increase of the satisfaction level after introducing the new system in the hospital

```
dat %>%  
  summarise(  
    mean_pre = mean(satisfaction_pre),  
    mean_post = mean(satisfaction_post)  
  )
```

```
mean_pre mean_post  
1  37.4625    75.25
```

One-way ANOVA

- Compares the mean of more than 2 groups in regards to the variation in the data
- Other names: one-factor ANOVA, between-subjects ANOVA, single-factor ANOVA
- One-way ANOVA is conducted when:
 - The dependent variable is numerical
 - The independent/grouping variable is categorical (>2 levels)
- Assumptions:
 1. Random sample
 2. The grouping and the outcome variables are independent of each other
 3. Normality - the outcome variable is normally distributed in each group
 4. Homoscedascity (equal variance) - the variance between groups is approximately normal

- Examples:
 - Comparing the systolic blood pressure between patients clinic A, B, and C
 - Comparing the BMI of doctors, pharmacists, and dentists in a hospital
- One-way ANOVA has an additional steps in the analysis - if one-way ANOVA is significant, post-hoc test should be conducted
- So:
 - One-way ANOVA: is there significant difference between the groups?
 - Post-hoc test: which group pairwise comparison is significant?
- Post-hoc test consists of doing several independent t-test for each pairwise comparison
- Since the independent t-tests are conducted several times on the same dataset, bonferroni adjustment is needed to adjust the p value (to avoid false positive)

One-way ANOVA: Analysis

- Background: we want to compare the recovery time between those receiving drugs A, B, and C
- Steps:
 1. State the hypothesis
 2. Check the assumptions
 3. Interpretation
 4. Conclusion

- Step 1:
 - H_0 : There is no difference in mean recovery time between patients receiving drugs A, B, and C
 - H_A : There is a difference in the mean recovery time for at least one of the drug groups

Packages

```
library(foreign)
```

```
library(knitr)
```

```
library(ggplot2)
```

```
library(dplyr)
```

```
library(car)
```

Read the data

```
dat <- read.spss("Data/anova.sav", to.data.frame = TRUE)
```

First 6 rows of the data

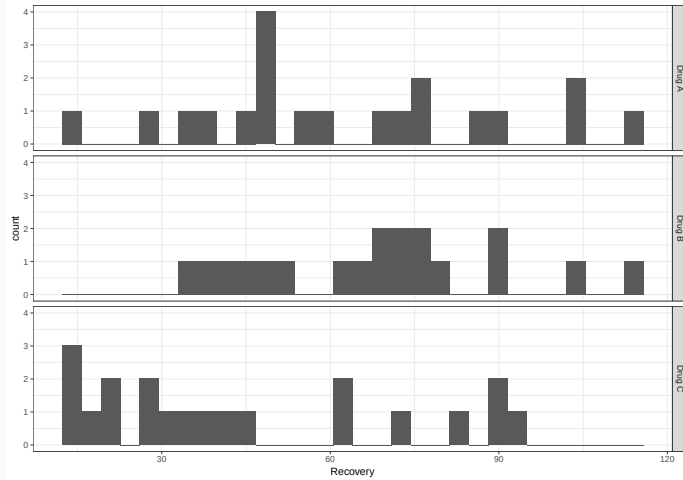
```
head(dat) %>% kable()
```

ID	Group	Recovery
1	Drug A	89
2	Drug A	104
3	Drug A	44
4	Drug A	57
5	Drug A	47
6	Drug A	76

- Step 2:

1. Normality - does not looks normal, but, let's assume normal

```
ggplot(dat, aes(Recovery)) +  
  geom_histogram() +  
  facet_grid(rows = vars(Group)) +  
  theme_bw()
```



- Step 2:
 2. Equal variance: $p = 0.25$ (equal variance is assumed)

Levene test

```
leveneTest(dat$Recovery, dat$Group, center = "mean")
```

Degrees of Freedom	F-statistic	p-value
2	1.422	0.25

- Step 3: Interpret one-way ANOVA:
 - $p \text{ value} < 0.05$
 - There is a difference in the mean recovery time for at least one of the drug groups
 - So, post-hoc test is indicated

```
ano.equal <- aov(Recovery ~ Group, dat)
summary(ano.equal)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Group	2	5691	2845.6	4.3	0.0183 *
Residuals	56	37059	661.8		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- Step 3: Interpret post-hoc test:
 - There is a significant difference between drug B and C after bonferroni adjustment

```
library(emmeans)
emmeans(ano.equal, pairwise ~ Group, adjust = "bonferroni")
```

Group	emmean	SE	df	lower.CL	upper.CL
Drug A	62.5	5.75	56	51.0	74.1
Drug B	68.2	5.90	56	56.3	80.0
Drug C	45.1	5.75	56	33.6	56.6

Confidence level used: 0.95

```
ano_res$contrasts
```

contrast	estimate	SE	df	t.ratio	p.value
Drug A - Drug B	-5.61	8.24	56	-0.680	1.0000
Drug A - Drug C	17.45	8.13	56	2.145	0.1089
Drug B - Drug C	23.06	8.24	56	2.798	0.0211

P value adjustment: bonferroni method for 3 tests

- Step 4: Conclusion:
 - There is a significant difference between the mean recovery time in at least one of the drug groups
 - Drug C had a significantly faster recovery time compared to Drug B
 - The rest of the pairwise drug comparisons were not significant